

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

ASSESSMENT TOOL 1 – CONCEPT MAPPING

Course Code	CSA0314	Course Title	Data Structures
CO Assessed	CO1	Assessment	Assessment Tool 1: Concept Mapping
Weightage	35% of CIA	Total Marks	100
CO4 Statement	<i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL2, BL3)</i>		
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Section/Batch	Slot C / 2026	Date	06-10-2026

Group 2 **Code of Conduct**

I, Dinakaran.S, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts	
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure	
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence	
Clarity	10%	Concept map is clear, neat,	Generally clear with minor	Clarity is reduced;	Poorly presented and hard to	

		visually structured, and easy to interpret	readability issues	difficult to interpret in parts	understand	
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File System Hierarchy – Suitable Non-Linear Data Structure

Topic: Data Structures

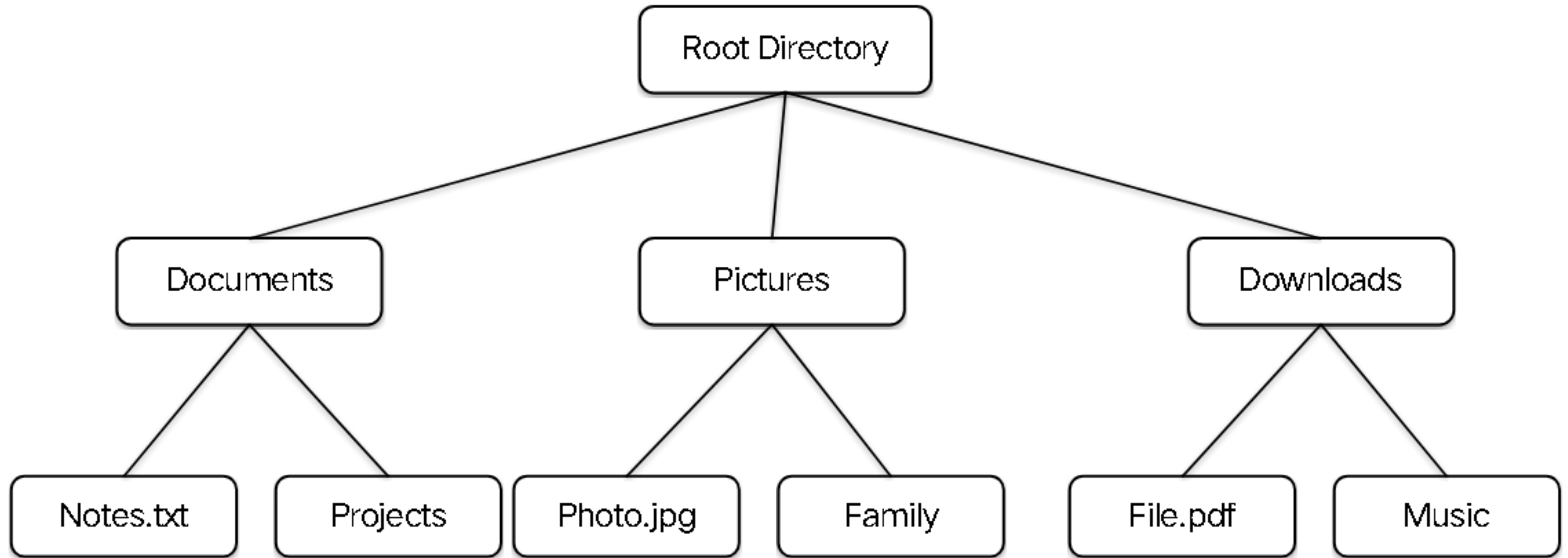
A file system organizes directories and subdirectories hierarchically.

Suitable data structure: TREE

Why Choose a Tree?

- A tree is a non-linear data structure.
- It represents parent – child relationships naturally.
- The root can represent the main directory.
- Subdirectories become child nodes.
- Files can be stored as leaf nodes.
- Each directory can contain multiple subdirectories and files.

File System as a Tree



Advantages of Using a Tree

1. Natural hierarchy – matches the structure of directories.
2. Easy navigation – move from parent directory to child directory.
3. Efficient organization – related files stay under the same directory.
4. Scalable – new folders and files can be added easily.
5. Searching – tree traversal methods can be used to locate files.
6. Clear relationships – parent and child directories are easy to identify.

Conclusion

The TREE is the most suitable non-linear data structure for a hierarchical file system.

It directly represents:

Root → Directories → Subdirectories → Files

Therefore, a tree provides a simple, organized, and scalable way to manage files and directories.